

by designing the Julia programming language and making it open source. He feels that while many like-minded people saw the potential early on, most did not believe that the two language problem—having two different languages for performance and ease of use — could be solved. He believes that he is still fighting that battle for mindshare, but has come a long way!

Shah says, “As projects grow, one has to often also build out organisational capacity for more contributors. These organisations are not well defined in the traditional sense and are challenging to govern and maintain. We think the Julia community has done immense work along these lines as it has grown from a small seed of four creators to thousands of contributors now.”

As of now, Shah continues to work on both Julia and Circuitscape. At Julia Computing, he is actively working with a talented team in the field of differentiable programming, which combines the areas of programming languages, scientific computing and machine learning.

Powering research in vaccines and drugs for diseases

Says Shah, “The work we are doing will make it possible to combine our model based understanding of science with the data-rich world we have entered over the last decade. Today, Julia is powering research in vaccines and drugs for diseases (including COVID-19), in climate science, new battery technology, and so many interesting applications. I feel blessed to be at the centre of all this, working on technologies that truly will impact the world at large.”

During the early days, when they started seeing industry beginning to use Julia, it became clear to the co-founders that in order to make the users successful, they had to set up a company. Shah says, “Six of us came together to found the company —

Alan Edelman, Jeff Bezanson, Stefan Karpinski, Deepak Vinchhi and Keno Fischer—along with me. The company was set up with the clear goal of providing the best technology for Julia users to develop, deploy and scale their Julia programs.

“Towards that, we built three products — the free JuliaPro distribution for individual users, JuliaTeam for enterprise users, and JuliaRun for deploying and scaling data science and AI codes. In addition, we also created JuliaAcademy.com and open sourced all our training materials.”

He says that none of the founders want the company to move away from its roots of innovation and research. Julia Computing collaborates with a number of partners to build cutting-edge domain specific technologies. An example of that is its partnership with Pumas-AI for pharmaceutical modelling and with Carnegie Mellon University for improving battery technology. It also maintains an active relationship with the Julia Lab at MIT for research into Julia, as well as with NumFocus for the open source Julia project. JuliaCon 2020 was the largest JuliaCon ever, with almost 29,000 attendees, and the third largest number of visitors were from India.

Infinite curiosity and an insatiable mind is what is needed

As someone who has been massively successful in this field, Shah has some sound advice to give to youngsters. He says, “Contributing to open source is a great way to build a career and a CV. This is because your software is then out there, and your contributions are visible publicly. In the pre-Internet days, you were limited to your college in finding mentors. With the Internet and open source, the world is your playground. We have seen so many amazing students from India participate through Google Summer of Code, Julia Season of Contributions,

and Major League Hacking. Many of these students have gone on to become long-term contributors in the Julia community.”

He advises picking an area or project that motivates us, where we will be able to work hard and long hours. Skillsets in languages and technologies can be learned with so many amazing online resources. One must have infinite curiosity, an insatiable mind, and the ability to persevere and work on something for years.

Talking about how India is doing and how the open source landscape has shaped up in the country, Shah says, “A friend once said that India is a country of downloaders — we only downloaded open source software but never uploaded any. I think this is changing in a big way. We see a huge number of contributions from India in the Julia ecosystem. Julia is one of the few big open source projects to have a major contribution from India. I also see many students participating in open source conferences, holding workshops, and interacting online.”

Leverage that mass in IT to build an open source movement in India

Shah feels there is a lot of room for improvement, though. Indian colleges and universities can teach the open source philosophy, and integrate their curriculum with open source contributions. “Our IT industry has millions of programmers. At any given time, so many programmers are on the bench, in between projects. Why not leverage that mass to build an open source movement in India, and put India on the map for open source software? A lot can be learnt from how Google Summer of Code is organised, and replicated at a larger scale in India,” concludes Shah. 

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